

Alexander C. Myrick

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EDUCATION

August 2026

- **Current Student at Emory University, Atlanta, GA (2024 – Ongoing)**
Completing B.S. in Computer Science & English minor
Emory University GPA: 4.0 / 4.0
- **Student at the University of Pittsburgh, Pittsburgh, PA (2023-2024)**
Honors College
Formerly completing B.S. in Computer Science & English minor
University of Pittsburgh GPA: 4.0 / 4.0

RESEARCH EXPERIENCE

- **REU CAAR: Combinatorics, Algorithms, and AI for Real Problems**
Location: University of Maryland, College Park
Funding: National Science Foundation (NSF)
Mentor: Dr. Laxman Dhulipala, University of Maryland, College Park
Project Title: I/O-efficient Parallel Algorithms in Theory and Practice
May 2026 – Ongoing
 - ☐ Currently developing and implementing algorithms with provably low I/O complexity for the multi-SSD external memory setting
 - ☐ Creating a user-friendly, highly parallel library intended to support operations on extremely large, locally distributed data (many times the typical capacity of DRAM)
- **NSF Directorate for Computer Science and Engineering (CISE) REU (Honors Thesis, Undergraduate Research Assistant)**
Advisor: Dr. Li Xiong, Emory University
Funding: NSF
Project Title: Toward Adversarially Robust and Privacy-Preserving EHR Systems: Data Complexity and New Training Paradigms
August 2026 – Ongoing
 - ☐ Currently developing novel time-series membership inference attacks for sepsis detection models
 - ☐ Working on derivations for principled membership inference attacks against generic sequence models

- **Undergraduate Research Assistant, Assured Information Management and Sharing Lab (AIMS)**
Advisor: Dr. Li Xiong, Emory University
 August 2025 – August 2026
 - ☐ Previously developed differentially private training techniques for multimodal machine learning models
 - ☐ Worked on developing privacy techniques for electronic health record data in fine-tuned models
- **Summer Undergraduate Research Experience (SURE), Atlanta, GA**
 Location: Emory University
 Funding: Emory University
Advisor: Dr. Joon-Seok Kim, Emory University
 May 2025 – August 2025
 - ☐ Worked on an original proposal regarding computational feasibility in agentic systems and surrogate modeling to reduce system load
 - ☐ Drafted a research paper that is being actively worked on
- **Undergraduate Research Assistant, AI and Simulation Lab**
Advisor: Dr. Joon-Seok Kim, Emory University
 December 2024 – May 2025
 - ☐ Worked on a project involving Large Language Model supervision in agentic simulations and machine learning

PRESENTATIONS/PUBLICATIONS

- **Multi-SSD Programming can be Fast and Ergonomic** (*presentation, REU-CAAR*)
 - ☐ Gave a short (20 minute) talk on multi-SSD programming work completed throughout the program and its potential applications
- **Data Privacy Literature Review Talks** (*presentation, AIMS Lab*)
 - ☐ Provided two long-form (45-60 minutes) presentations on relevant data privacy literature to other lab members and advisor
- **Surrogate Modeling for Agentic Systems** (*poster, Emory SURE*)
 - ☐ Participated in and created poster for long-form session on agentic simulation work
- **Multi-SSD Programming can be Fast and Ergonomic** (*paper, title in progress*)
 - ☐ To be submitted to ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) 2027

WORK EXPERIENCE

- **Montessori School of Columbia**, Columbia, SC
Chess Teacher, Afterschool supervisor, Summer Program Instructor, January 2022 – June 2023
 - ☐ Provided chess lessons, boards, puzzles, and coaching for all grade levels weekly (separated into two age groups, 1-4 and 5-8)
 - ☐ Organized school-wide tournaments complete with prizes and trophies
 - ☐ Selected as a guest instructor for a week-long summer program that expanded on the fundamentals of chess taught during the school year
 - ☐ Assisted afterschool workers with overseeing the students, cleanup, and check-out procedures

SKILLS

- Proficient in C, C++, Python, Java, Bash Scripting, SQL, and Assembly
- Experience in computer science subareas including artificial intelligence/machine learning, simulations, data privacy, and both practical and theoretical parallel algorithms
- Experience in functional programming ideas and languages (primarily Haskell)
- Tournament Chess Player, experience in state/national level tournaments, formerly held state-level title (2023)
- Experienced piano performer; 27 recitals (including 4 university-level juries and a university-affiliated concert) and one paid-entry performance at Carnegie Hall, NY completed
- Certified in First Aid, CPR, and AED use

ACHIEVEMENTS

- **Academic, National Merit Scholarship Foundation** – National Merit Scholarship Finalist
- **Academic, United States Department of Education** – Presidential Scholars Candidate
- **Extracurricular, South Carolina Independent School Association** – 1st Place/117 in Spring 2023 SC State Independent Chess Championship
- **Extracurricular, South Carolina Independent School Association** – 2nd Place/171 in Fall 2022 SC State Independent Chess Championship